

```
import numpy as np

array = np.random.randint(1, 100, 16)
array
```

```
array([71, 19, 58, 61, 55, 10, 61, 73, 28, 61, 71, 77, 39, 13, 46, 37])
```

```
array.mean()
```

```
np.float64(48.75)
```

```
np.percentile(array, 25)
```

```
np.float64(34.75)
```

```
np.percentile(array, 50)
```

```
np.float64(56.5)
```

```
np.percentile(array, 75)
```

```
np.float64(63.5)
```

```
np.percentile(array, 100)
```

```
np.float64(77.0)
```

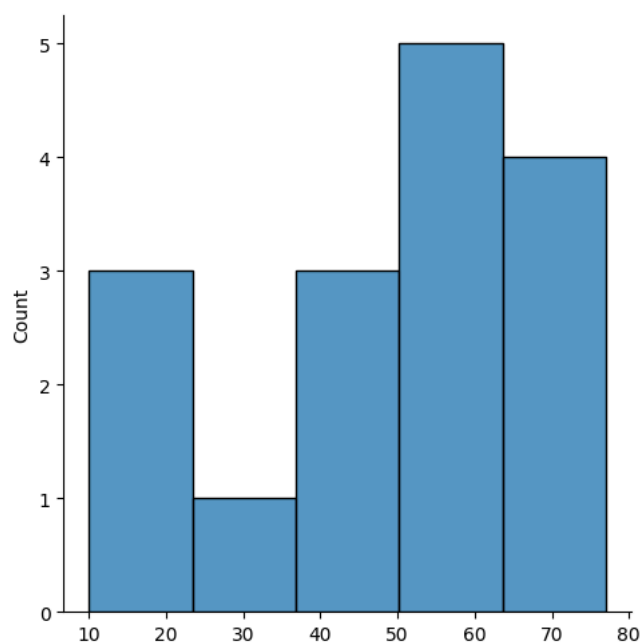
```
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur
```

```
lr, ur = outDetection(array)
lr, ur
```

```
(np.float64(-8.375), np.float64(106.625))
```

```
import seaborn as sns
import matplotlib.pyplot as plt
```

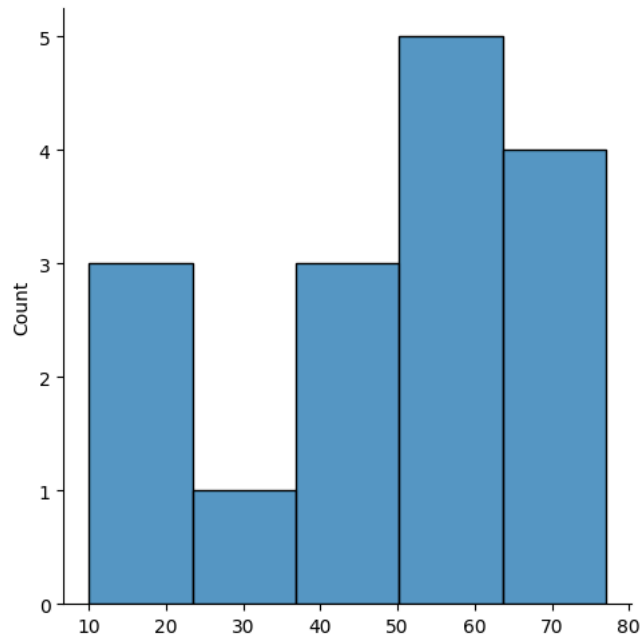
```
sns.displot(array)
plt.show()
```



```
new_array = array[(array > lr) & (array < ur)]
new_array
```

```
array([71, 19, 58, 61, 55, 10, 61, 73, 28, 61, 71, 77, 39, 13, 46, 37])
```

```
sns.displot(new_array)  
plt.show()
```



```
lr1, ur1 = outDetection(new_array)  
lr1, ur1final_array = new_array[(new_array > lr1) & (new_array < ur1)]  
final_array
```

```
(np.float64(-8.375), np.float64(106.625))
```

```
final_array = new_array[(new_array > lr1) & (new_array < ur1)]  
final_array
```

```
array([71, 19, 58, 61, 55, 10, 61, 73, 28, 61, 71, 77, 39, 13, 46, 37])
```

```
sns.displot(final_array)  
plt.show()
```

